Surface-Potential-Based Modeling of MOS Capacitors Including Non-Ideal Effects for SiC CMOS Processes

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Summary

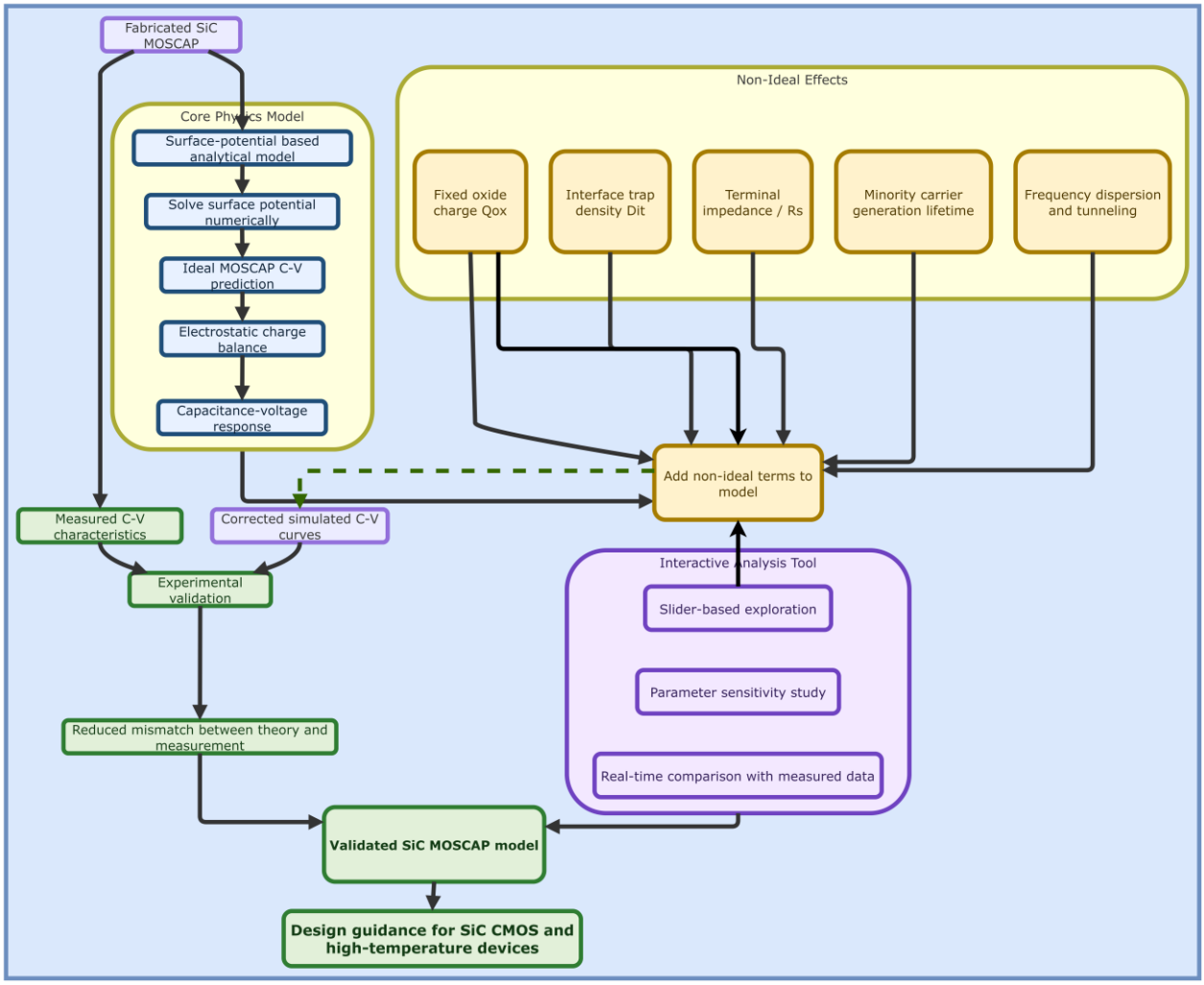
The metal-oxide-semiconductor capacitor (MOSCAP) is one of the most extensively studied and well-understood devices in semiconductor physics, especially for silicon (Si). However, practical MOSCAPs, especially non-silicon ones often exhibit non-ideal behaviors due to fabrication processes, oxide interfaces, material properties, measurement-related anomalies, interface trapping, and material defects etc is still not explored thoroughly. Efforts have been made to model MOSCAP for both Silicon and SiC based devices which generally ignore non-ideal scenarios. In this study, A surface potential based analytical model has been developed to accommodate different non-ideal effects. This study illustrates how interface trap density, fixed oxide charge, terminal impedance, and minority carrier generation lifetime influence capacitance-voltage (C-V) characteristics and overall device behavior. An interactive visualization tool has been developed to simulate the model. The visualization tool provides valuable insights into the impact of these non-ideal effects and can aid in the analysis, design, and optimization of SiC-based MOS devices.

Motivation

Silicon-based MOSCAPs are crucial for understanding underlying device physics and are foundational for integrated circuit (IC) design using standard CMOS processes. Although the MOSCAP has been extensively studied for Si, Silicon Carbide (SiC) based CMOS processes have recently emerged to enable operation at high temperatures and high power. Unlike Si, SiC exhibits numerous defects, as well as interface and measurement-related anomalies due to interface trap density, fixed oxide charge, terminal impedance, and minority carrier generation lifetime, low intrinsic career concentration in wide bandgap materials which result is frequency dispersion, interface trap-related deviations in C-V characterization. Recently Model for low voltage MOSFET including C-V behavior for CMOS process has been introduced however only account the interface trap density while ignoring others.[1] While this serve an excellent for IC design but due to the BSIM4 based model architecture it calculated C-V empirically. To accurately analyze these errors, there is a critical need for physics based models that accurately capture SiC-specific physics—to advance circuit simulation and progress of SiC CMOS process development in this study, we developed a surface potential-based model for a SiC MOSCAP including the non-ideal behaviors and validated it against measured results.

Results

The results show that the initial measured and analytical characteristics deviate due to the presence of fixed oxide charge, terminal impedance, and interface trap density. However, when these non-ideal effects are incorporated into the analytical model, the discrepancy is significantly reduced, and the simulated results closely match the measured data. This strong agreement validates both the proposed model and the fabricated SiC MOSCAP. These findings demonstrate that accounting for non-ideal effects is essential for accurate modeling and inclusion in the process design kit (PDK) to be considered in the circuit design and simulation of SiC-based MOSCAPs and CMOS process that is being developed for high temperature and harsh environment.



Summary of the proposed Surface potential based MOSCAP model with Non-Ideal Behavior of SiC Device

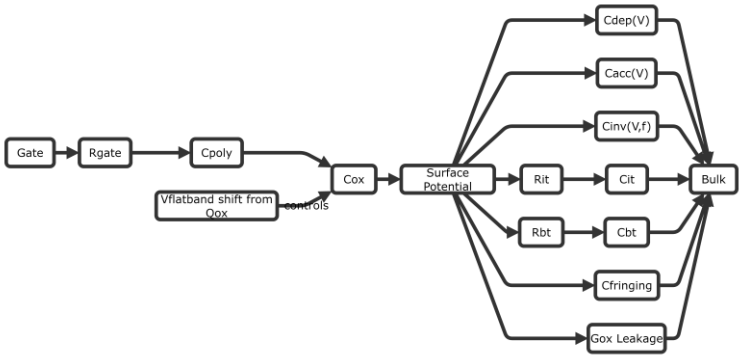


Figure 1: Schematics representation of the Model

OVERVIEW:

PRELIMINARY RESULTS

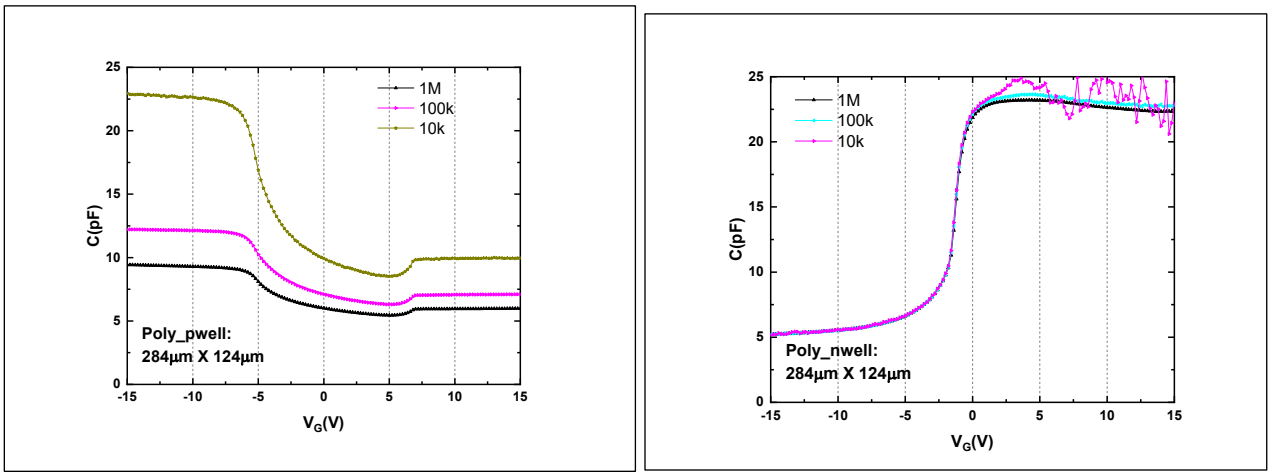


Figure 2: Characterization result Poly to Pwell and Poly to Nwell MOS cap in different frequency fabricated from SiC CMOS process

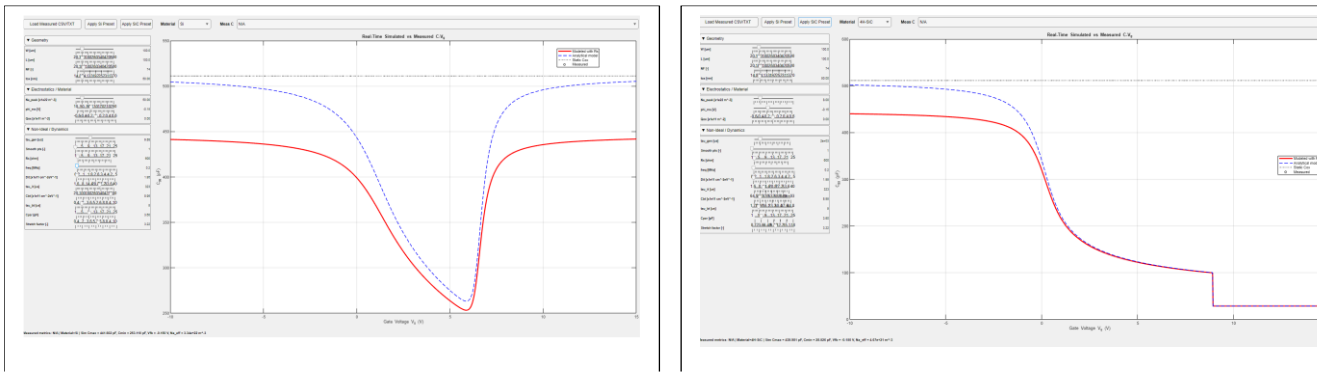


Fig. 3: Interactive visualization tool for non-ideal effect visulization

A surface potential based MOSCAP model has been developed including non-ideal behavior of SiC MOSCAP and compared it with measured data to validate the model accuracy. An interactive visualization tool also been developed for understanding the impact of non-ideal behavior for MOSCAP in real time.

Conclusion:

This study bridges the gap between theoretical models and practical device behavior in SiC MOSCAPs. By developing an analytical surface potential model validated by experimental data, we demonstrated that incorporating non-ideal effects—such as fixed oxide charge, terminal impedance, and interface traps—is critical for accurate device characterization. Supported by a newly developed visualization tool, these findings provide a vital foundation for advancing the design and optimization of high-performance SiC CMOS processes.

Reference:

A. S. M. K. Hasan, M. M. Hossain, M. Z. Islam, M. Majharul Islam, S. Ahmed and H. Alan Mantooth, "Temperature Scaling and C–V Modeling of SiC Low-Voltage MOSFETs for IC Design," in *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 13, no. 1, pp. 335-347, Feb. 2025, doi: 10.1109/JESTPE.2024.3482971.